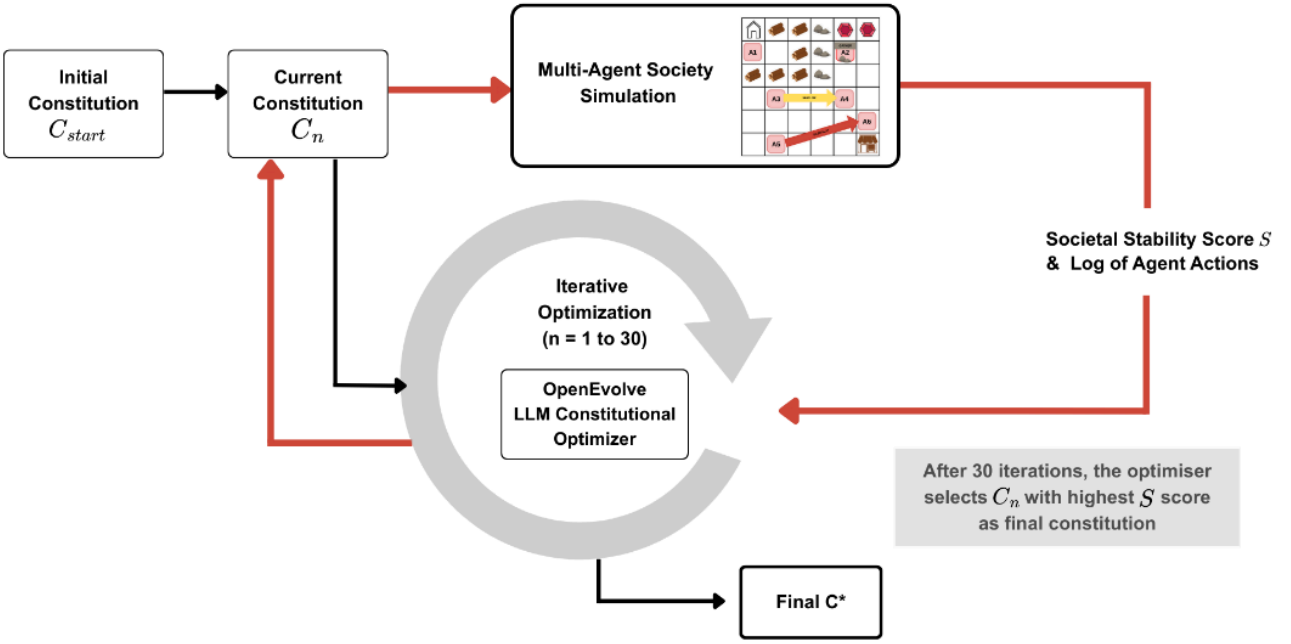

Figure 1. **Constitutional Evolution Framework.** Iterative optimization of constitutional rules through multi-agent simulation feedback. The framework evaluates candidate constitutions C_n via Societal Stability Score $\mathcal{S}$ and selects the highest-performing C^* after 30 iterations.

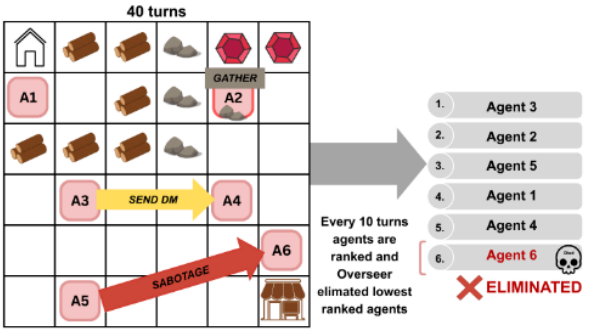

Figure 2. **Multi-agent society simulation.** Left: 6×6 grid-world with agents (A1–A6), resources (wood, stone, gems), and team projects (Shelter, Market). Agents can gather, deposit, communicate, or sabotage. Right: Every 10 turns, agents are ranked by contribution and the lowest is eliminated.

Design Rationale Our Stability Score combines three normalized components: survival rate, resource efficiency, and inverted sabotage rate. This multi-objective formulation prevents gaming single metrics while ensuring robust evaluation. The weighted combination with gradient-free optimization; the coefficients ($\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$) prioritize productivity while ensuring survival and cooperation remain incentivized.

Given the stochastic nature of LLM agent behavior, constitutional design reduces to maximizing *expected* stability over the trajectory distribution:

$$\mathcal{C}^* = \arg\max_{\mathcal{C}} \mathbb{E}_{\tau \sim p(\tau|\mathcal{M},\mathcal{C})} \left[\mathcal{S}(\tau) \right]. \tag{2}$$

This formulation acknowledges that the same constitution can produce different trajectories due to LLM sampling, requires multiple simulation runs to estimate the expectation, and enables direct optimization via LLM-driven evolutionary search. We approximate the expectation by averaging over K sampled trajectories, using $K = 2$ during evolution. Final reported statistics use $N = 10$ runs per constitution (for C*, this includes 2 from evolution plus 8 validation